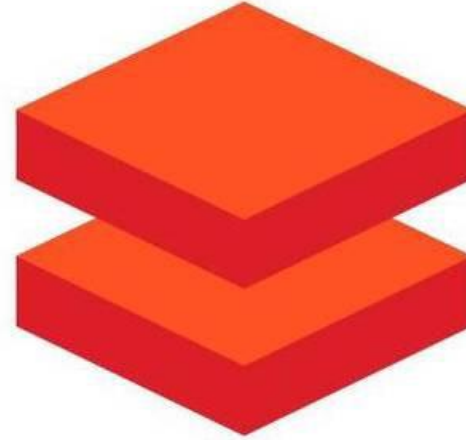




Spark Introduction

Agenda

- Limitations of Hadoop
- History of Spark
- Spark
- Hadoop vs. Spark
- Spark Components
 - Spark Core, Spark SQL, Spark Streaming, Spark Mlib, GraphX
- Spark Architecture
- Applications & Use Case

Limitations of Hadoop

- Slow Processing Speed
- No real-time data processing
- No Caching
- Lengthy Line of Code
- Support for Batch Processing Only

History of Spark

- 2009 → Started as a Project at UC Berkley AMP Lab
- 2010 → Open Sourced
- 2013 → Spark became an Apache Top Level Project
- 2014 → Used by data bricks for large scale datasets

Hadoop vs. Spark

Hadoop vs. Spark

Hadoop

Spark

Hadoop vs. Spark

Hadoop

- Processing data using MapR in Hadoop is slow

Spark

- Spark processes data 100 times faster than MapReduce

Hadoop vs. Spark

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- Spark processes data 100 times faster than MapReduce
- Performs both batch processing and real-time processing of data

Hadoop vs. Spark

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- Java (More lines of Code)

Spark

- Spark processes data 100 times faster than MapReduce
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- Scala (Fewer lines of Code)

Hadoop vs. Spark

Hadoop

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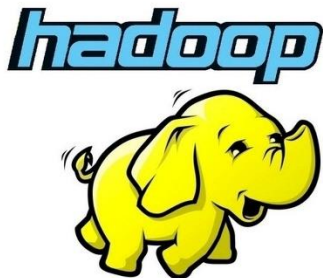
Spark

- Spark processes data 100 times faster than MapReduce
- Performs both batch processing and real-time processing of data
- Scala (Fewer lines of Code)
- Caches the data in memory

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Spark

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Apache Spark

- Open Source Framework
- Data Processing Engine to process data in real-time
- In-memory data processing
- Reading data from source(hdfs:// or s3://) is time consuming. So after you read data from the source and apply all the common operations, cache it if you are going to reuse the data.

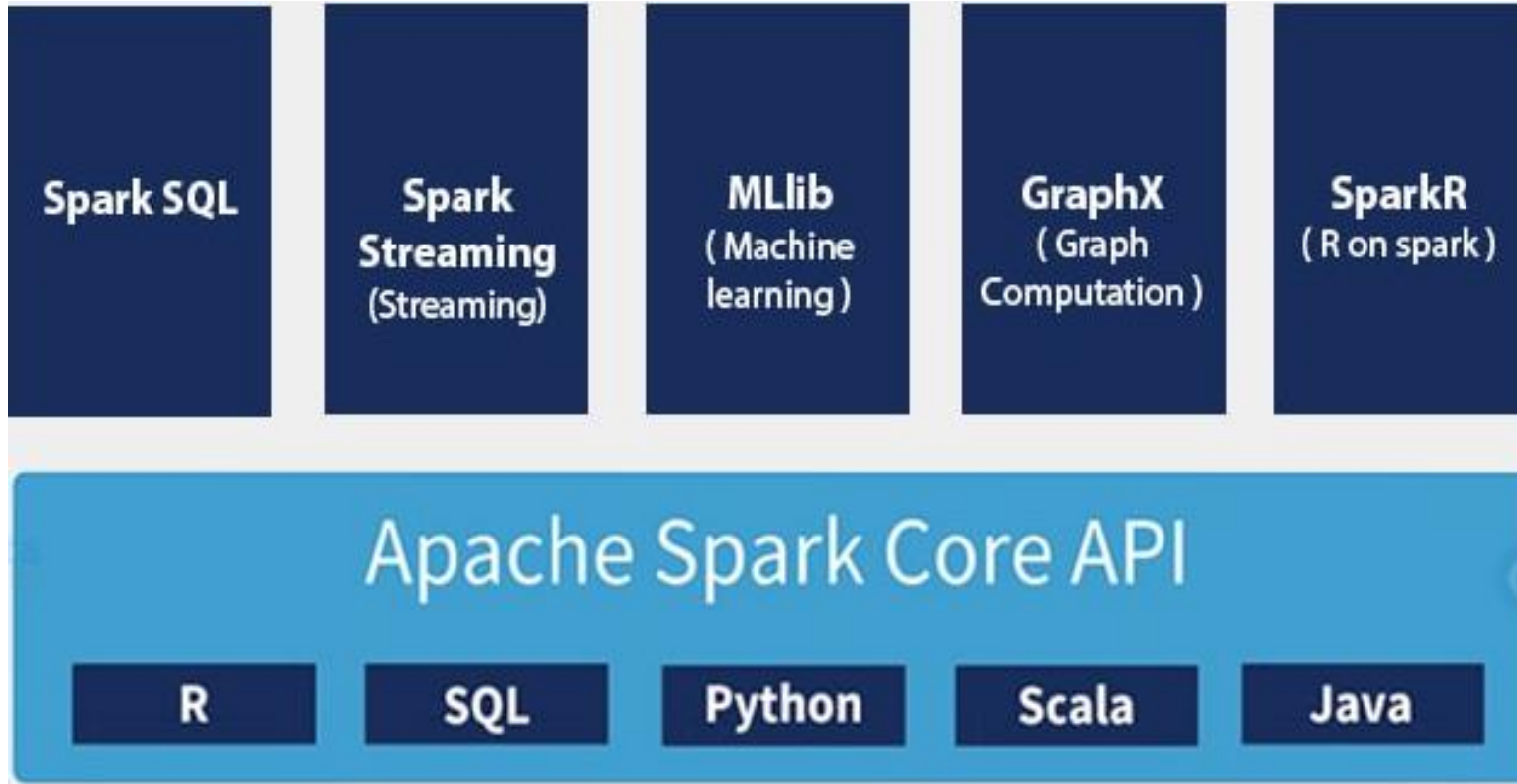
Hard disk / RAM



Features of Spark

- Fast Processing (RDDs → saves time in reading and writing)
- In memory Computing (data stored in RAM)
- Flexible (supports multiple languages → Java, Python, R, Scala)
- Fault Tolerance (RDDs → temp datasets)
- Better Analytics (ML, Complex analytics)

Components of Spark



Spark Core

- Base Engine for large-scale parallel and distributed data processing
 - Memory Management
 - Fault Recovery
 - Scheduling, Distributing and monitoring jobs
 - Interacting with storage systems

Spark SQL

- Spark SQL framework component is used for structured and semi-structured data processing
 - CSV
 - JSON
 - JDBC
 - Data Frame APIs
 - Data source APIs

Spark Streaming

- Spark Streaming is a lightweight API that allows developers to perform batch processing and real-time processing of data
- Provides,
 - Secure
 - Reliable
 - Fast Processing

Spark MLlib

- MLlib low level machine learning library that is simple to use
- Scalable
- Compatible
 - Clustering
 - Classification
 - Recommendation

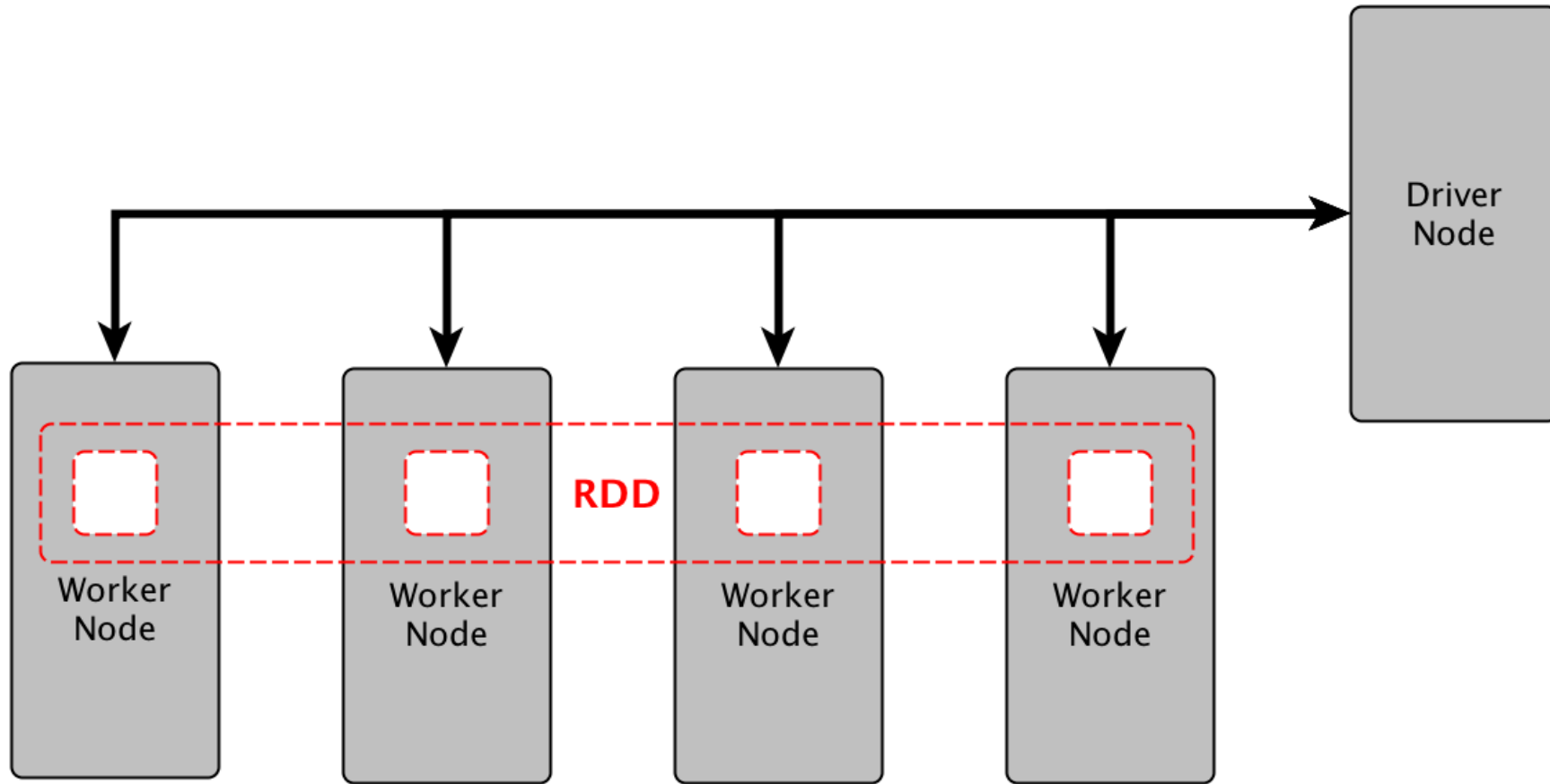
GraphX

- GraphX is spark's own computation Engine
 - Interactive Graph
 - EDA
 - Provides tool for ETL

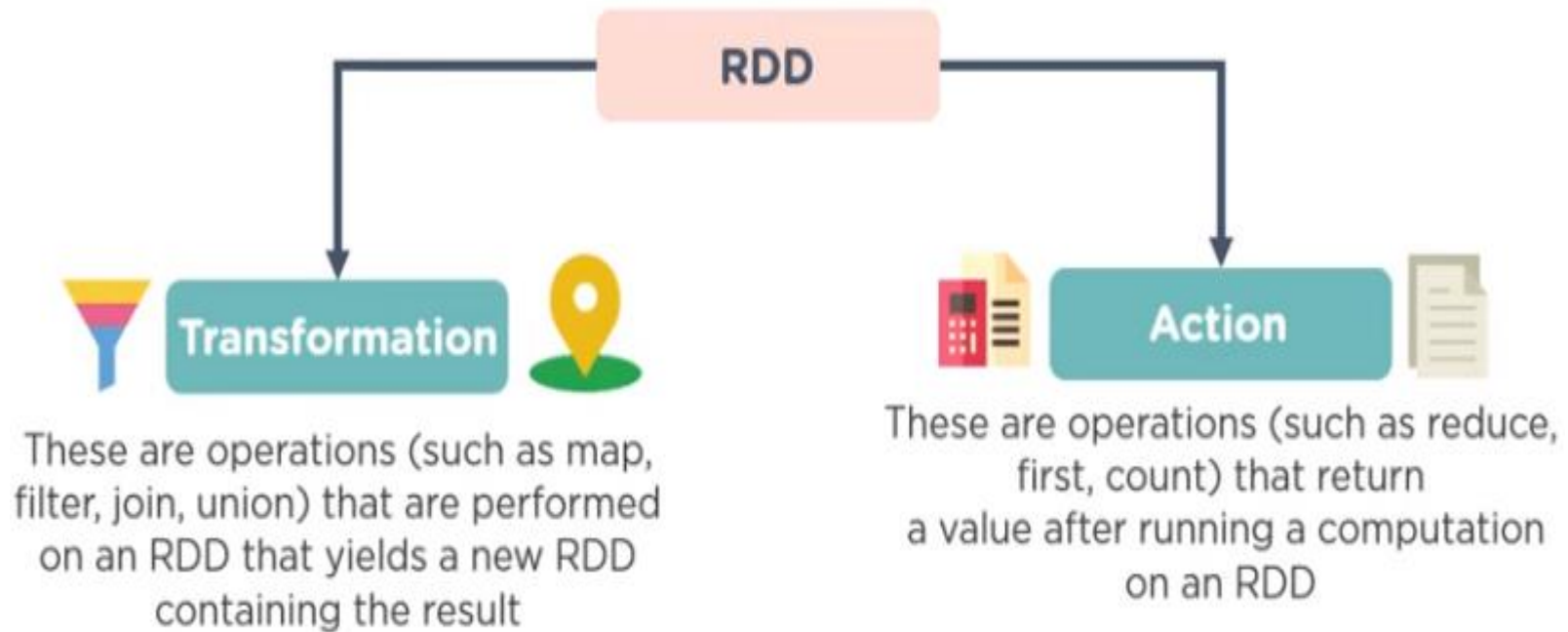
RDDs (Resilient Distributed Dataset)

- RDD(Resilient Distributed Dataset) is immutable distributed collection of objects. RDD is a logical reference of a dataset which is partitioned across many server machines in the cluster.
- RDD could come from any data source, e.g. text files, a database via JDBC, etc.

RDDs (Resilient Distributed Dataset)

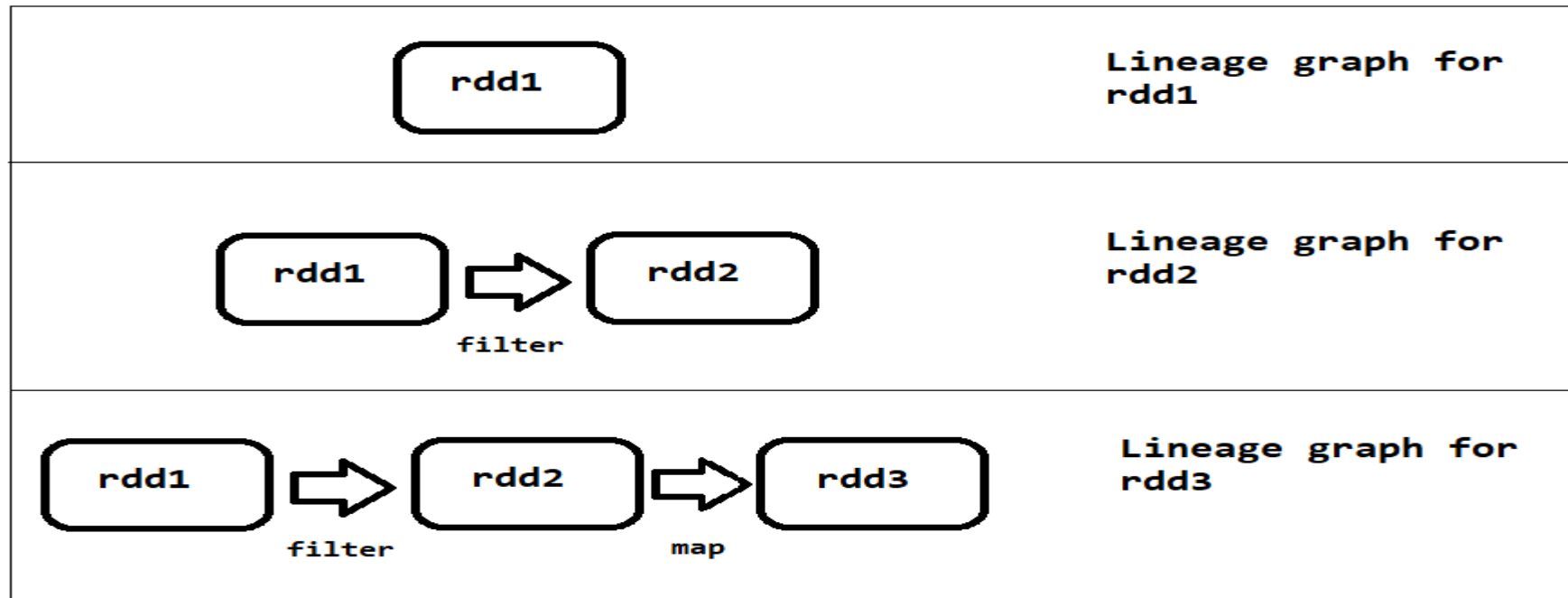


RDDs (Resilient Distributed Dataset)

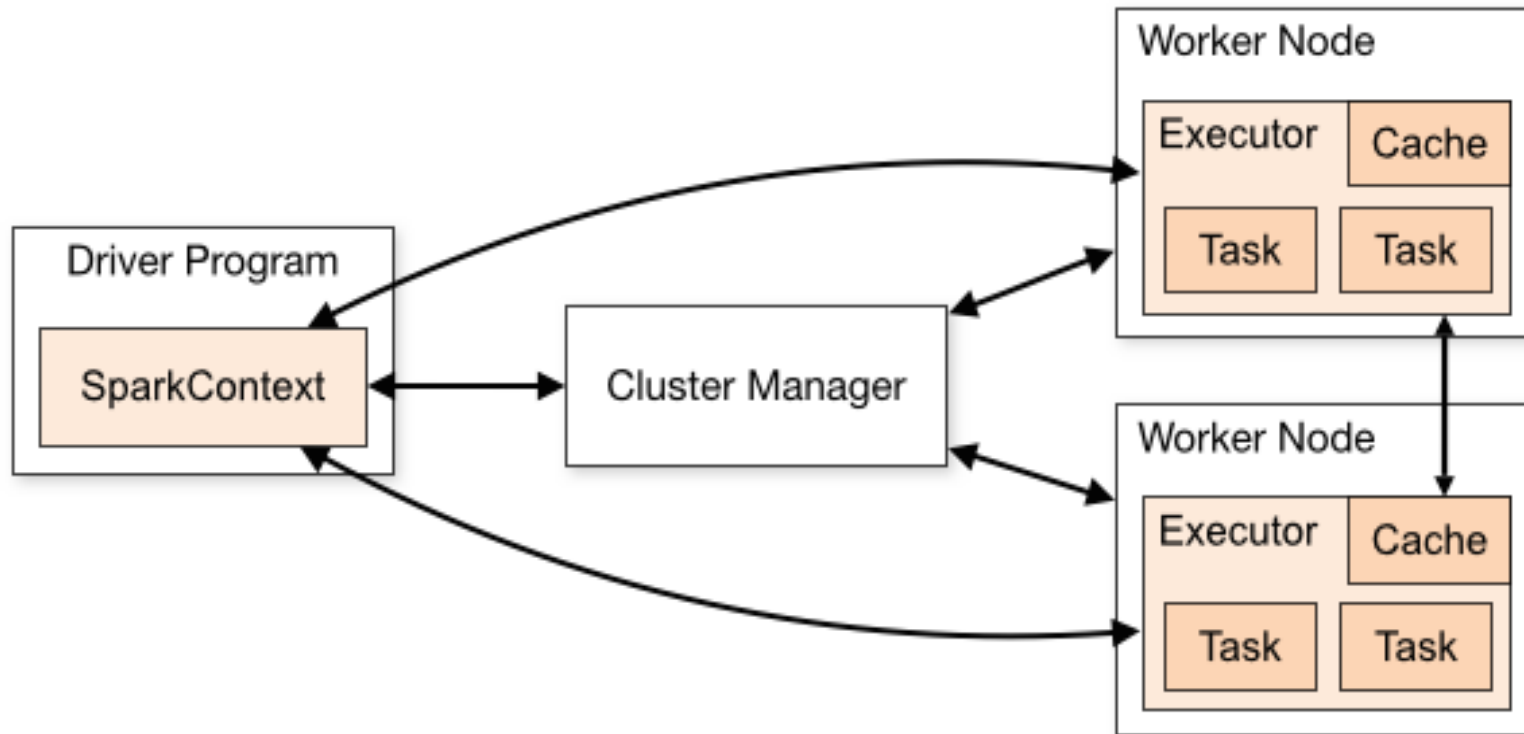


DAG(Directed Acyclic Graph)

- **DAG Scheduler** is the scheduling layer of Apache Spark that implements **stage-oriented scheduling**
- It Transforms a **logical execution plan**



Spark Architecture



Spark Cluster Managers

- Apache Standalone Mode
- MESOS (open source project to manage cluster)
- Hadoop Yarn
- Kubernetes (automating deployment)

Applications & Use Case

- Fraudulent Transactions
- Analyse patterns
- Browsing history
- Analyse health data and diagnose
- Netflix

Thank You !!!

Questions??